

Players can change the color of a wolf's collar even if they're not its owner

MC-197241 Items

Status	Resolved / Fixed
Affected	1.16.1, 1.16.3, 1.16.4, 20w46a, 20w48a, 20w51a (+14 more)
Fixed in	23w12a
Reported	2020-08-06

STEPS TO REPRODUCE

1. Launch Minecraft.
2. Create or enter a Creative world.
3. Obtain any color dye.
4. Summon a wolf that is not owned by you by using `/summon minecraft:wolf ~ ~ ~ {Owner:"Notch"}`.
5. Use the dye on the summoned wolf.
6. Observe that the wolf's collar color changes even though you are not its owner.

OBSERVED BEHAVIOUR

A player who is not the owner of a tamed wolf can still interact with it using a dye. If the dye is a different color from the wolf's current collar, the interaction succeeds and the collar color changes to the selected dye color. The dye is also consumed for players who are not in Creative mode.

The wolf's `mobInteract` logic checks that the wolf is tamed and that the selected dye differs from the current collar color, but it does not consistently restrict this collar-color change to the player returned by `isOwnedBy(player)`. As a result, any player can recolor the collar of another player's tamed wolf. This behavior is specific to wolves; cats correctly verify ownership before allowing equivalent interactions.

EXPECTED BEHAVIOUR

Only the wolf's owner should be able to change the color of its collar using dye. When a player who does not own the tamed wolf attempts to apply dye, the collar color should remain unchanged and the interaction should not consume the dye. The owner should still be able to recolor the collar normally.

ENVIRONMENT

Minecraft Java Edition Versions: 1.16.1, 1.16.3, 1.16.4, 20w46a, 20w48a, 20w51a, 21w06a, 21w08a, 21w08b, 21w11a, 1.17, 1.17.1, 21w41a, 1.18.1, 1.18.2, 22w19a, 1.19, 1.19.2, 1.19.3, 1.19.4 (includes snapshots)

ORIGINAL DESCRIPTION

The Bug:

Players can change the color of a wolf's collar even if they're not its owner.

It's important to note that this concept is exclusive to wolves and cannot be seen with cats. This issue isn't present with cats because the `isOwnedBy()` method is called appropriately with this entity, therefore preventing other players from being able to interact with them.

Steps to Reproduce:

- Obtain any color dye and summon a wolf that is not your owner by using the command provided below.
`/summon minecraft:wolf ~ ~ ~ {Owner:"Notch"}`
- Attempt to change the color of the wolf's collar by using the dye.
- Take note as to whether or not players can change the color of a wolf's collar even if they're not its owner.

Observed Behavior:

Players can change the color of a wolf's collar even if they're not its owner.

Expected Behavior:

Players would not be able to change the color of a wolf's collar if they're not its owner.

Code Analysis:

Code analysis by Avoma can be found below.

The following is based on a decompiled version of Minecraft 1.19.2 using MCP-Reborn.

net.minecraft.world.entity.animal.Wolf.java

```
public class Wolf extends TamableAnimal implements NeutralMob {
    ...
    public InteractionResult mobInteract(Player player, InteractionHand interactionHand) {
        ItemStack itemstack = player.getItemInHand(interactionHand);
        Item item = itemstack.getItem();
        if (this.level.isClientSide) {
            ...
        } else {
            if (this.isTame()) {
                ...
                DyeColor dyecolor = ((DyeItem)item).getDyeColor();
                if (dyecolor != this.getCollarColor() && this.isOwnedBy(player)) {
                    this.setCollarColor(dyecolor);
                    if (!player.getAbilities().instabuild) {
                        itemstack.shrink(1);
                    }
                }
                return InteractionResult.SUCCESS;
            }
            ...
        }
    }
}
```

If we look at the above class, we can see that there are two checks that are carried out before allowing the color of a wolf's collar to be changed. One of these checks is to see if the wolf is tamed and the other is to see if the player is holding a dye that's not already the same color as the wolf's collar at the time of the interaction. If these two requirements are met, the color of the wolf's collar can be changed. The game doesn't check if the player carrying

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-197241>